

Ibn Tofail University

Analysis V

Problem Set II

Exercise 1:

Study the existence of a limit at $(0, 0)$ for the following functions:

$$f(x, y) = \frac{x}{y}; \quad f(x, y) = \frac{1 - \cos(xy)}{y^2}; \quad f(x, y) = \frac{(x^2 + y^2)^2}{x^2 - y^2}$$

$$f(x, y) = \frac{2x^2 + xy}{\sqrt{x^2 + y^2}}; \quad f(x, y) = \frac{\sin(xy)}{|x| + |y|}; \quad f(x, y) = \begin{cases} \frac{|xy|^\alpha}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}, \text{ where } \alpha > 0$$

Correction

Exercise 2:

Study the limit at $(0, 0)$ of the two-variable numerical function f defined by:

$$f(x, y) = \frac{\cos x - \cos y}{x - y}$$

Correction

Exercise 3:

Study the existence of a limit at the indicated point for the following functions:

$$f(x, y, z) = \frac{xyz}{x + y + z} \text{ at } (0, 0, 0); \quad f(x, y, z) = \frac{x + y}{x^2 - y^2 + z^2} \text{ at } (2, -2, 0)$$

Correction

Exercise 4:

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function defined by

$$f(x, y) = \begin{cases} 2y & \text{if } x \geq y^2 \\ \frac{2x}{y} & \text{if } |x| < y^2 \text{ and } y \neq 0 \\ -2y & \text{if } x \leq -y^2 \end{cases}$$

Study the continuity of f on $\mathbb{R}^2$.

Correction**Exercise 5:**

Study the continuity on $\mathbb{R}^2$ of the functions defined in Exercises 2 and 3.

Correction**Exercise 6:**

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function defined by

$$f(x, y) = \begin{cases} \frac{|x|^\alpha \cdot |y|^{2\alpha}}{|x|^\alpha + |y|^{2\alpha}} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Determine the values of α for which f is continuous at $(0, 0)$.

Correction